

The Geometry of Linear Algebra

Lecture 03: Row, Column, and Matrix Pictures

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Abstract

These notes review the fundamental concepts for viewing a system of linear equations from three distinct but interconnected perspectives: the row picture, the column picture, and the matrix form. We will use a simple 2×2 system to illustrate how these geometric and algebraic interpretations provide a deeper understanding of the solution, with connections to machine learning applications.

Contents

1	Introduction: Why Linear Algebra Matters in Machine Learning	3
1.1	Real-World Machine Learning Connections	3
2	Key Concept: Linear System of Equations	3
2.1	General Definition	3
2.2	Our Central Example	4
3	Solving the System Algebraically	4
4	The Row Picture: Intersection of Lines	4
4.1	Graphing the Lines	4
5	The Column Picture: A Linear Combination of Vectors	5
5.1	Formulating the Vector Equation	5
6	The Matrix Form: $Ax = b$	6
6.1	Formulating the Matrix Equation	6
6.2	Matrix-Vector Multiplication as Linear Combination	7
7	Teaching Focus: The Why-What-How Framework	7
8	Future Directions in Linear Algebra for ML	8
9	Summary and Key Insights	8

1 Introduction: Why Linear Algebra Matters in Machine Learning

🔍 Why-What-How Framework

Why Machine Learning models often rely on linear operations: transformations, projections, and optimization. Many algorithms (e.g., linear regression, neural networks, PCA) involve solving systems of linear equations. Understanding these systems builds intuition for how models "learn" from data.

What A linear system of equations is a collection of equations that describe relationships among variables using linear combinations. These systems can be represented compactly using matrices and vectors.

How We represent these relationships in matrix form and use linear algebra operations to find solutions efficiently. The geometric interpretations provide crucial intuition for understanding model behavior.

1.1 Real-World Machine Learning Connections

🏠 Machine Learning Connection

Linear Regression: Predicting house prices based on features (area, bedrooms, location) involves solving $A^T A x = A^T b$ to find optimal weights.

Neural Networks: Each layer performs linear transformations followed by non-linear activations: $z = Wx + b$.

PCA (Principal Component Analysis): Finds directions of maximum variance through eigenvalue decomposition of covariance matrices.

Recommendation Systems: Matrix factorization techniques break down user-item interaction matrices into lower-dimensional representations.

2 Key Concept: Linear System of Equations

2.1 General Definition

A linear system consists of multiple linear equations with common variables:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

In compact matrix form:

$$A\mathbf{x} = \mathbf{b}$$

where:

- A : $m \times n$ matrix of coefficients (a_{ij})
- $\mathbf{x}$: column vector of unknowns $[x_1, x_2, \dots, x_n]^T$
- $\mathbf{b}$: column vector of constants $[b_1, b_2, \dots, b_m]^T$

2.2 Our Central Example

We'll use a simple 2×2 system throughout this lecture to build intuition:

$$\begin{cases} 2x + y = 3 \\ x - 2y = -1 \end{cases}$$

In matrix form: $\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$

3 Solving the System Algebraically

Step 1: Isolate a variable. From the second equation ($x - 2y = -1$):

$$x = 2y - 1$$

Step 2: Substitute. Substitute into the first equation ($2x + y = 3$):

$$2(2y - 1) + y = 3$$

Step 3: Solve for y .

$$4y - 2 + y = 3$$

$$5y - 2 = 3$$

$$5y = 5$$

$$y = 1$$

Step 4: Back-substitute to find x .

$$x = 2(1) - 1$$

$$x = 1$$

Solution

The unique solution to the system is **(x, y) = (1, 1)**. This is the only pair of values that simultaneously satisfies both equations.

4 The Row Picture: Intersection of Lines

💡 Core Idea

The row picture considers each equation (or "row") in the system individually. In a 2D system, each linear equation represents a straight line on the xy -plane. The solution to the system is the single point where all the lines intersect.

4.1 Graphing the Lines

- **Line 1:** $2x + y = 3$ (or $y = -2x + 3$)
Points: (0, 3) and (1, 1)

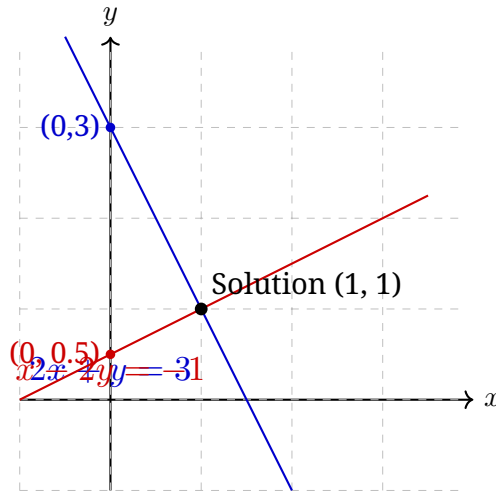


Figure 1: The Row Picture: Two lines intersecting at the solution point (1,1).

- **Line 2:** $x - 2y = -1$ (or $y = \frac{1}{2}x + \frac{1}{2}$)
Points: (0, 0.5) and (1, 1)

Machine Learning Connection

Higher Dimensions: In 3D, each equation represents a plane, and the solution is their intersection line or point. In n-dimensions (common in ML), each equation represents a hyperplane.

Support Vector Machines: Use the concept of finding the optimal separating hyperplane between classes.

Decision Boundaries: Linear classifiers create decision boundaries that are intersections of half-spaces defined by linear equations.

5 The Column Picture: A Linear Combination of Vectors

Core Idea

The column picture reinterprets the system as a single vector equation. We group the coefficients of each variable into column vectors. The goal is to find the correct "linear combination" of these column vectors that produces the vector on the right-hand side.

5.1 Formulating the Vector Equation

$$x \begin{pmatrix} 2 \\ 1 \end{pmatrix} + y \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

Key vectors:

- Column Vector 1: $\vec{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ (coefficients of x)
- Column Vector 2: $\vec{v}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ (coefficients of y)

- Target Vector: $\vec{b} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ (constants)

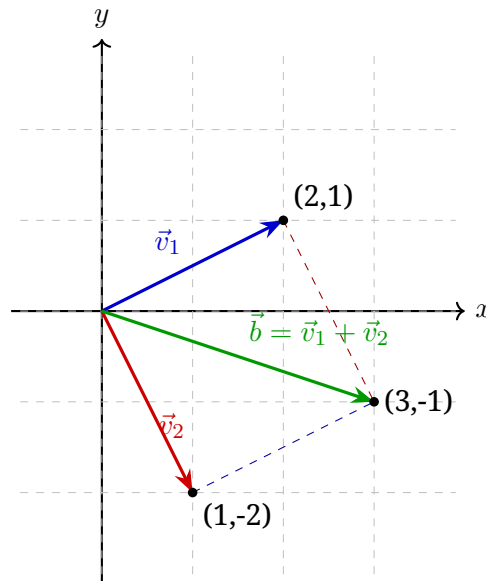


Figure 2: The Column Picture: The sum of vector $\vec{v}_1$ and vector $\vec{v}_2$ produces the target vector $\vec{b}$.

Machine Learning Connection

Feature Space: In ML, each column vector represents a feature, and we're trying to find the right combination of features to match the target output.

Word Embeddings: In NLP, words are represented as vectors, and linear combinations can capture semantic relationships.

Portfolio Optimization: In finance, finding the right mix of assets (column vectors) to achieve target returns.

6 The Matrix Form: $Ax = b$

Core Idea

The matrix form provides the most compact and powerful way to represent a linear system. It encapsulates all coefficient information into a single matrix A , the unknowns into a vector x , and the constants into a vector b .

6.1 Formulating the Matrix Equation

- **Coefficient Matrix (A):** $\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$
- **Vector of Unknowns (x):** $\begin{pmatrix} x \\ y \end{pmatrix}$
- **Constant Vector (b):** $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$

$$\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

6.2 Matrix-Vector Multiplication as Linear Combination

The product $A\mathbf{x}$ is exactly the linear combination from the column picture:

$$A\mathbf{x} = x \cdot (\text{first column of } A) + y \cdot (\text{second column of } A)$$

Machine Learning Connection

Neural Networks: The forward pass through a dense layer is exactly matrix-vector multiplication: $\mathbf{z} = W\mathbf{x} + \mathbf{b}$.

Linear Transformations: Matrices represent linear transformations that preserve vector space structure.

Efficient Computation: Modern hardware (GPUs) is optimized for matrix operations, making deep learning feasible.

7 Teaching Focus: The Why-What-How Framework

Table 1: Comprehensive View of Linear Systems in Machine Learning

Focus	Mathematical Perspective	Geometric Interpretation	Machine Learning Application
Why	To model linear relationships between variables	Find intersection points/spaces that satisfy all constraints	Model training, prediction, feature combination
What	System $A\mathbf{x} = \mathbf{b}$ with matrix A , vector $\mathbf{x}$, vector $\mathbf{b}$	Intersection of lines/planes/hyperplanes	Learning weights that map inputs to outputs
How	Gaussian elimination, matrix inverses, least squares	Geometric constructions and vector operations	Gradient descent, backpropagation, SVD

8 Future Directions in Linear Algebra for ML

Advanced Topics Preview

- **Matrix Inverses:** Solving $\mathbf{x} = A^{-1}\mathbf{b}$ for efficient solutions
- **Eigenvalues/Eigenvectors:** Understanding matrix behavior and dimensionality reduction (PCA)
- **Singular Value Decomposition (SVD):** Matrix factorization for recommendation systems
- **Vector Spaces and Basis:** Understanding the "space" where data lives
- **Linear Regression:** Solving $A^T A \mathbf{x} = A^T \mathbf{b}$ for optimal fits
- **Convex Optimization:** Finding global minima of quadratic loss functions

9 Summary and Key Insights

Key Takeaway

- **Row Picture:** Algebraic problem $\rightarrow$ Geometric intersection problem
- **Column Picture:** Finding the right linear combination of feature vectors
- **Matrix Form:** Compact representation enabling efficient computation
- All three perspectives describe the same mathematical reality but provide different insights
- The ability to switch between these views is crucial for deep understanding

Learning Outcomes

After this lecture, you should be able to:

- Translate between row, column, and matrix representations of linear systems
- Visualize 2D and 3D systems geometrically
- Understand how linear algebra forms the foundation for machine learning
- Recognize the connection between matrix operations and linear combinations
- Appreciate why multiple perspectives are valuable in mathematics